Dimensionality Reduction and Market Structure Analysis of the Indian Equity Market Using Principal Component Analysis

1 Introduction and Research Context

The Indian equity market is a complex financial system comprising hundreds of publicly traded companies whose returns exhibit significant co-movement. The NIFTY 500 index represents a broad cross-section of the Indian economy and includes firms from diverse sectors such as information technology, energy, financial services, and manufacturing.

Analyzing the joint behavior of such a large number of stocks presents challenges related to dimensionality, redundancy, and interpretability. Principal Component Analysis (PCA) provides a statistical framework to reduce dimensionality by identifying a smaller number of uncorrelated components that capture the dominant patterns of variation in asset returns.

2 Data Description

2.1 Stock Universe Selection

To ensure consistency in the application of PCA, this study fixes the stock universe at a single point in time. The set of stocks included in the NIFTY 500 index as of the year 2005 is selected as the base universe. Only those stocks that have continuous price data over the entire study period and are not delisted are retained. This results in a fixed universe of N stocks, where $N < 500$.

This approach avoids changes in the dimensionality of the data matrix over time and allows for meaningful comparison of PCA results across different periods.

2.2 Time Period and Frequency

The study uses daily data over a 15-year period from 2005 to 2020, corresponding to approximately $T \approx 3{,}700$ trading days. Daily frequency is chosen as it balances data availability, economic interpretability, and the reduction of market microstructure noise present in high-frequency data.

2.3 Return Construction

For each stock i, the daily log-return at time t is computed using adjusted closing prices:

$$r_{t,i} = \frac{1}{\sqrt{\Delta t}} \ln\left(\frac{P_{t,i}^{\text{adj}}}{P_{t-1,i}^{\text{adj}}}\right)$$

where $P_{t,i}^{\text{adj}}$ accounts for corporate actions such as dividends, stock splits, and bonus issues.

The resulting return matrix is denoted as

$$\mathbf{R} \in \mathbb{R}^{T \times N}.$$

3 Problem Statement

Although the NIFTY 500 consists of hundreds of individual stocks, their returns are not independent. Shared exposure to macroeconomic conditions, sectoral trends, and market-wide events leads to strong correlations across assets.

The central problem addressed in this study is whether the high-dimensional return space of the Indian equity market can be approximated by a smaller number of latent components, and how the interpretation of these components changes over time.

4 Methodology

4.1 Return Standardization

The raw return series of different stocks exhibit varying levels of volatility due to differences in firm size, sectoral risk, and trading activity. If PCA is applied directly to unscaled returns, stocks with higher volatility may dominate the principal components, leading to results driven by scale rather than genuine co-movement.

To avoid this issue, the return series of each stock are standardized prior to analysis. For each stock i, the standardized return $\tilde{r}_{t,i}$ is computed as:

$$\tilde{r}_{t,i} = \frac{r_{t,i} - \mu_i}{\sigma_i},$$

where μ_i and σ_i denote the sample mean and sample standard deviation of returns for stock i, respectively.

This transformation ensures that each stock contributes equally to the PCA in terms of variance. Consequently, the extracted principal components reflect common patterns of movement across stocks rather than differences in individual stock volatility.

4.2 Principal Component Analysis

Let $\tilde{\mathbf{R}} \in \mathbb{R}^{T \times N}$ denote the standardized return matrix, where each column corresponds to a stock and each row corresponds to a trading day.

The sample covariance matrix of returns is computed as:

$$\mathbf{C} = \frac{1}{T-1}\tilde{\mathbf{R}}^\top \tilde{\mathbf{R}},$$

where $\mathbf{C} \in \mathbb{R}^{N \times N}$ captures the linear dependence structure among the N stocks.

Principal Component Analysis is performed by solving the eigenvalue decomposition:

$$\mathbf{C}\mathbf{w}_j = \lambda_j \mathbf{w}_j, \quad j = 1, \ldots, N,$$

where λ_j denotes the j-th eigenvalue and $\mathbf{w}_j$ is the corresponding eigenvector.

The eigenvectors are ordered such that:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N.$$

Each eigenvector $\mathbf{w}_j \in \mathbb{R}^N$ represents a principal component loading vector, indicating the contribution of each stock to that component.

4.3 Factor Scores and Matrix Approximation

The factor scores are obtained by projecting the standardized returns onto the principal component directions:

$$\mathbf{F} = \tilde{\mathbf{R}}\mathbf{W},$$

where:

- $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k] \in \mathbb{R}^{N \times k}$ is the matrix of the first k eigenvectors (factor loadings),
- $\mathbf{F} \in \mathbb{R}^{T \times k}$ contains the factor scores (principal component time series).

Using the first k principal components, the standardized return matrix can be approximated as:

$$\tilde{\mathbf{R}} \approx \mathbf{F}\mathbf{W}^\top.$$

This low-rank approximation captures the dominant sources of variance in the data while discarding higher-order components that primarily reflect noise.

4.4 Subperiod Analysis

To study the stability and interpretation of principal components over time, the 15-year sample is divided into three non-overlapping subperiods:

- 2005–2010
- 2010–2015
- 2015–2020

PCA is performed separately for each subperiod using the same fixed stock universe.

4.5 Sector-Wise Loading Analysis

To facilitate economic interpretation of the principal components, stocks are grouped according to their sector classifications. For a given principal component j, the contribution of each sector is quantified by computing the average absolute loading of stocks within that sector.

Let $\mathcal{S}_m$ denote the set of stocks belonging to sector m, and let $|\mathcal{S}_m|$ denote the number of stocks in that sector. The sector-wise average absolute loading for principal component j is defined as:

$$L_{m,j} = \frac{1}{|\mathcal{S}_m|} \sum_{i \in \mathcal{S}_m} |w_{ij}| \,,$$

where w_{ij} represents the loading of stock i on the j-th principal component.

This measure captures the typical magnitude of exposure of stocks within a sector to a given principal component. Sectors with larger values of $L_{m,j}$ are interpreted as contributing more strongly to the corresponding component.

5 Expected Results and Outputs

The primary outputs of this thesis include:

- Eigenvalue spectra and scree plots for the full sample and each subperiod.
- Cumulative variance explained by the leading principal components.
- Tables showing the number of components required to explain a substantial proportion (approximately 70–85%) of total variance.
- Heatmaps of factor loadings illustrating sectoral dominance.
- Comparative analysis demonstrating how the interpretation of leading components shifts across time periods.

6 Discussion

Changes in the sectoral interpretation of leading principal components are expected across different subperiods. For example, one period may exhibit dominance by information technology stocks, while another may be driven by energy or financial sectors. Such shifts reflect changes in market conditions and economic structure rather than instability of the PCA methodology.

7 Conclusion

This study demonstrates that despite the high dimensionality of the Indian equity market, a relatively small number of principal components can capture a significant portion of return

variation. Moreover, the dominant sources of market variance evolve over time, highlighting the dynamic nature of market structure.

The findings underscore the usefulness of PCA as an exploratory tool for understanding market-wide co-movements while acknowledging its limitations in providing fixed economic interpretations.

8 Limitations and Future Work

The analysis is limited by the use of a fixed stock universe and linear dimensionality reduction. Future work may extend this study by incorporating rolling-window PCA, alternative factor models, or nonlinear dimensionality reduction techniques.